Use Cases and User Stories

Dormfix — Hostel Complaint & Maintenance Management System

1. Actors

Actor	Description
Resident	A hostel student who reports and tracks maintenance issues
Warden	The staff member who reviews, prioritizes, and resolves complaints

2. Use-Case Diagram



3. Use-Case Descriptions

UC-01: Register Account

- **Actor:** Resident
- **Precondition:** User does not already have an account

- **Main Flow:**
    1. Resident opens the registration page
    2. Enters full name, registration number, room number, email, password
    3. System validates all fields and uniqueness of email/registration number
    4. System creates the account with role "resident" and logs the user in
- **Alternate Flow:** If email or registration number already exists, system shows an error and does not create the account
- **Postcondition:** Resident account exists and is authenticated

UC-02: Login

- **Actor:** Resident, Warden
- **Precondition:** User has a valid account
- **Main Flow:**
    1. User enters email and password
    2. System verifies credentials
    3. System issues a JWT and redirects to the appropriate dashboard based on role
- **Alternate Flow:** Invalid credentials → system shows a generic "invalid email or password" error (no hint as to which field is wrong)

UC-03: Submit Complaint

- **Actor:** Resident
- **Precondition:** Resident is logged in
- **Main Flow:**
    1. Resident opens "Create Complaint"
    2. Enters a description and optionally attaches an image
    3. System auto-fills name, room number, and registration number from the resident's profile
    4. System generates a unique complaint ID and sets status to SUBMITTED
- **Alternate Flow:** Description too short → validation error shown; invalid image type/size → upload rejected

UC-04: View My Complaints

- **Actor:** Resident
- **Main Flow:** Resident opens the dashboard and sees a list/history of only their own complaints with current status and overdue indicator

UC-05: View Complaint Details

- **Actor:** Resident, Warden
- **Main Flow:** User opens a specific complaint and sees full details (description, image, timestamps, status)
- **Alternate Flow (Resident):** Attempting to open another resident's complaint ID directly is rejected with a 403 Forbidden response

UC-06: Submit Feedback

- **Actor:** Resident

- **Precondition:** Complaint status is RESOLVED and no feedback has been submitted yet
- **Main Flow:** Resident rates the resolution (1–5) and optionally adds a comment
- **Alternate Flow:** Feedback attempted before resolution, or a second time on the same complaint → rejected

UC-07: Logout

- **Actor:** Resident, Warden
- **Main Flow:** User clicks logout; token is cleared locally and user is redirected to login

UC-08: View All Complaints

- **Actor:** Warden
- **Main Flow:** Warden opens their dashboard and sees every complaint from every resident, newest first by default

UC-09: Filter / Sort Complaints

- **Actor:** Warden
- **Main Flow:** Warden applies filters (status, overdue, room) and/or changes the sort order; the list updates accordingly

UC-10: Reorder Priority

- **Actor:** Warden
- **Main Flow:** Warden manually moves a complaint up or down in the priority list; the system persists the new order
- **Note:** This is a manual decision only — the system does not calculate or suggest priority automatically

UC-11: Update Complaint Status

- **Actor:** Warden
- **Main Flow:** Warden changes a complaint's status following the allowed transitions (SUBMITTED → UNDER_PROGRESS → RESOLVED, or SUBMITTED → RESOLVED directly)
- **Alternate Flow:** An invalid transition (e.g. RESOLVED → SUBMITTED) is rejected by the server regardless of what the frontend sends

UC-12: View Feedback

- **Actor:** Warden
- **Main Flow:** Warden opens a resolved complaint and sees the resident's rating and comment, if submitted

UC-13: View Dashboard Statistics

- **Actor:** Warden
- **Main Flow:** Warden sees aggregated counts (total, by status, overdue) and a simple resolved-per-day chart

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4. User Stories with Acceptance Criteria

US-01: As a resident, I want to register with my hostel details so that I can start submitting complaints.

- **Acceptance Criteria:**
  - Given valid unique details, the account is created and I am logged in
  - Given a duplicate email or registration number, I see a clear error and no account is created

US-02: As a resident, I want to submit a complaint with a description and photo so that the warden understands the issue clearly.

- **Acceptance Criteria:**
  - Given a description of at least 5 characters, the complaint is created with status SUBMITTED
  - Given an image over 5MB or an unsupported format, the upload is rejected with a clear message

US-03: As a resident, I want to see only my own complaints so that my information stays private from other residents.

- **Acceptance Criteria:**
  - My dashboard never shows another resident's complaint
  - Directly requesting another resident's complaint by ID returns a 403 error

US-04: As a resident, I want to rate how my complaint was resolved so that I can give feedback on the maintenance process.

- **Acceptance Criteria:**
  - Feedback form only appears after status is RESOLVED
  - I cannot submit feedback twice on the same complaint

US-05: As a warden, I want to see all complaints in one place so that I don't rely on informal reporting channels.

- **Acceptance Criteria:**
  - All complaints from all residents appear, newest first by default
  - I can filter by status, overdue flag, and room number

US-06: As a warden, I want to manually reorder complaints so that I can prioritize based on my own judgment, not an automated score.

- **Acceptance Criteria:**
  - I can move any complaint up or down
  - The system never reorders complaints on its own

US-07: As a warden, I want the system to flag complaints overdue by more than 48 hours so that nothing gets forgotten.

- **Acceptance Criteria:**
  - The overdue flag is calculated from the server clock, not trusted from the browser
  - A resolved complaint is never shown as overdue